Problem Solving with C

LABORATORY MANUAL

Week 4

Semester: 2
Course Code: UE25CS151B
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Commands to execute Programs using gcc

1. Editing
    $gedit filename.c

2. Compile
    $ gcc -c filename.c

3. Linking
    $ gcc filename.o

4. Loading - Execute
    $ ./a.out (Linux Operating System)
    $ a.exe (Windows Operating System

Program 1: Banana Level

Write a C program that prints Floyd's Triangle. The program should prompt the user to enter an integer representing the number of rows in the triangle. The program should then print the triangle in the following format: (Use for loop)

If the user enters 4 then the output will be:
1
2 3
4 5 6
7 8 9 10

<u>Sample Output:</u>
Enter number of rows: 4
1
2 3
4 5 6
7 8 9 10

Program 2: Banana Level

A bank wants to implement a simple ATM security system.

When a customer inserts the ATM card, they must enter a 4-digit PIN to access their account. (Use while loop)

Rules:

1. The correct PIN is 1234.
2. The customer is allowed a maximum of 3 attempts.
3. If the user enters the correct PIN:
    - Display:
      "Access Granted!"
    - Stop further attempts.
4. If the entered PIN is incorrect:
    - Display:
      "Wrong PIN. Attempts left: X"
      (where X is the remaining number of attempts)
5. If the user fails all 3 attempts:
    - Display:
      "Card Blocked!"

<u>Sample output:</u>

Enter PIN: 1234
Access Granted!

Enter PIN: 123
Wrong PIN. Attempts left: 2
Enter PIN: 2891
Wrong PIN. Attempts left: 1
Enter PIN: 2781
Wrong PIN. Attempts left: 0
Card Blocked!

Enter PIN: 123
Wrong PIN. Attempts left: 2
Enter PIN: 234
Wrong PIN. Attempts left: 1
Enter PIN: 1234
Access Granted!

Program 3: Orange Level

Write a C program that performs the following tasks: (Use for loop)

Input:

1. A positive integer num (base number for multiplication table)
2. An integer limit (how far the table should go)
3. An integer threshold

Tasks:

Using a for loop:

1. Print the multiplication table of num up to limit in the format:
2. num x i = result
3. While generating the table, also:
    - Calculate the sum of all multiplication results.
    - Count how many results are greater than the given threshold.
    - Track the maximum result generated.

After printing the table, display:

- Total sum of results
- Number of results greater than threshold
- Maximum result in the table

<u>Sample output:</u>
Enter the base number: 3
Enter the limit: 20
Enter the threshold value: 30
Multiplication Table:
3 x 1 = 3

3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
3 x 11 = 33
3 x 12 = 36
3 x 13 = 39
3 x 14 = 42
3 x 15 = 45
3 x 16 = 48
3 x 17 = 51
3 x 18 = 54
3 x 19 = 57
3 x 20 = 60
----- Summary -----
Total Sum = 630
Results greater than 30 = 10
Maximum Result = 60

Program 4: Loan EMI Recovery Until Closure –Jack Level

A borrower has principal P. Every month:

- Interest added = P * r / 100
- EMI paid = E
- Penalty: if payment is less than interest for that month, add penalty ₹200.

Stop when loan closes (P <= 0) or month count exceeds N. (Use while loop)

Input: initial P, monthly rate r, EMI E, max months N

Tasks:

- Use while to update outstanding.
- Print month-wise outstanding (rounded to 2 decimals).
- Output total interest paid, total penalties, months taken, and if not closed within N, print remaining P.

<u>Sample output:</u>

Enter Principal (P): 10000
Enter Monthly Interest Rate (r in %): 2

Enter EMI (E): 2500
Enter Max Months (N): 6

Month-wise Outstanding:
Month 1 Outstanding = 7700.00
Month 2 Outstanding = 5354.00
Month 3 Outstanding = 2961.08
Month 4 Outstanding = 520.30
Month 5 Outstanding = 0.00

----- Summary -----
Total Interest Paid: 530.71
Total Penalties: 0.00
Months Processed: 5
Loan Status: CLOSED

Program 5: Smart Parking Lot Billing – Jack Level

A mall parking lot records vehicles one by one until the operator enters Vehicle Type = 0. (Use while loop)

Input per vehicle:

- Vehicle type: 1 = Bike, 2 = Car, 3 = SUV, 4 = Truck, 0 = Stop
- Hours parked (can be fractional, e.g., 2.5)

Rules:

- Bike ₹20/hr, Car ₹40/hr, SUV ₹60/hr, Truck ₹80/hr
- If hours > 6 → flat 15% discount on that vehicle's bill
- If total vehicles processed is a multiple of 10 → add ₹50 service charge to the *10th vehicle* bill (only that one)

Tasks:

- Use a while loop to process vehicles.
- Print each vehicle bill.
- At the end print: total vehicles, total revenue, max bill, min bill.

<u>Sample output:</u>
Smart Parking Lot Billing System
Enter Vehicle Type (1=Bike, 2=Car, 3=SUV, 4=Truck, 0=Stop)

Enter vehicle type: 1
Enter number of hours parked: 4
Bill for vehicle 1 = Rs. 80.00

```
Enter vehicle type (0 to stop): 0

----- Final Summary -----
Total Vehicles: 1
Total Revenue: Rs. 80.00
Maximum Bill: Rs. 80.00
Minimum Bill: Rs. 80.00


Smart Parking Lot Billing System
Enter Vehicle Type (1=Bike, 2=Car, 3=SUV, 4=Truck, 0=Stop)

Enter vehicle type: 2
Enter number of hours parked: 3
Bill for vehicle 1 = Rs. 120.00

Enter vehicle type (0 to stop): 1
Enter number of hours parked: 3
Bill for vehicle 2 = Rs. 60.00

Enter vehicle type (0 to stop): 0

----- Final Summary -----
Total Vehicles: 2
Total Revenue: Rs. 180.00
Maximum Bill: Rs. 120.00
Minimum Bill: Rs. 60.00
```